INSTITUTIONAL TECHNICAL SPECIFICATION & SYSTEM BRIEFING

BIOSECURE AI — SYSTEM OVERVIEW

Automated Multi-Face Recognition Attendance & Autonomous Biometric Drift Monitoring Platform

PROJECT ESSENCE — WHAT THE PROJECT IS:

BioSecure AI is an enterprise-grade, contactless attendance automation platform designed for universities, academic institutions, and organizations. It replaces manual roll calls (which waste 10–15 minutes of lecture time), eliminates magnetic/RFID swipe cards prone to proxy attendance ('buddy-punching'), and removes the physical queues and sanitary concerns of optical fingerprint scanners. By capturing or uploading a single classroom group photograph, the platform autonomously detects and aligns dozens of faces in parallel, extracts 512D deep embeddings, matches identities against enrolled student rosters in **under 1 millisecond on CPU**, logs attendance records to a PostgreSQL database, and proactively tracks biometric template aging across semesters.

HOW IT WORKS — THE 5-STAGE AUTOMATED INFERENCE PIPELINE

1. Multi-Face Capture Captures classroom photo via smartphone PWA window, webcam stream, or file upload.	**2. Detection & Align** RetinaFace locates 50+ faces simultaneously, extracting 5 landmarks for canonical alignment.	**3. ArcFace Encoding** Extracts 512-dimensional deep representations normalized to unit Euclidean sphere (||v|| = 1.0).	**4. In-Memory Match** NumPy C-BLAS computes Q · M^T batch dot product, matching full rosters in <1.0 ms on CPU.	**5. Log & Drift Engine** Logs 'Present' in PostgreSQL pgvector. Evaluates 3D Pose Gate and accumulates EWMA drift.

CORE CAPABILITIES & SYSTEM ARCHITECTURE

System Capability	Technical Implementation	Operational Impact & Performance Profile
Dense Group Recognition	RetinaFace ResNet50 + ArcFace Buffalo_L ONNX	Identifies 50+ students in parallel from a single wide-angle snapshot in <900 ms turnaround.
Sub-Millisecond Matching	NumPy BLAS Matrix Dot-Product (AVX2 SIMD)	<1.0 ms CPU execution for 1,000 students; fallback to Supabase pgvector HNSW RPC query.
Biometric Drift Engine	3D Pose Gate (|Yaw|≤25°, |Pitch|≤20°) + EWMA (α=0.30)	Continuously monitors facial template aging (hair, beard, weight) over multi-year degree programs.
Neural OCR ID Reader	RapidOCR (DBNet+SVTR ONNX) + OpenCV Multi-Pass	Extracts student Name, Roll ID, Program, Branch, and Email directly from physical ID cards in <1.2s.
Progressive Web App (PWA)	Service Worker (sw.js) + HTML5 capture='environment'	Direct smartphone camera launch (ChatGPT-style), offline app shell fallback, and zero app-store friction.
Class Roster Modes	Separate Single Class vs Combined Multi-Branch Mode	Enforces strict curriculum boundaries or evaluates joint auditorium cohorts dynamically.
Automated Alerting	Asynchronous SMTP Email Dispatcher + /admin/drift	Dispatches warning emails upon CRITICAL drift (≥0.25) with 1-click re-enrollment template reset.
Security & Privacy (RLS)	PostgreSQL Row-Level Security + Non-Invertible Vectors	Biometric embeddings cannot reverse-engineer faces; strict role separation (Instructor vs Admin).

TECHNOLOGY STACK & OPERATIONAL BENCHMARKS

AI & Computer Vision: InsightFace (Buffalo_L), RetinaFace, ArcFace 512D, RapidOCR (DBNet+SVTR ONNX), OpenCV.	**Backend & Persistence:** Python 3.10+, Flask Factory, Gunicorn WSGI, Nginx Proxy, PostgreSQL 16 + pgvector (HNSW Index).	**Client & Frontend:** Progressive Web App (PWA), Service Worker, Tailwind CSS Dark Theme, Lucide Icons, WebRTC.	**Key Benchmarks:** • Vector Match: <1.0 ms CPU • End-to-End: <1.0 s • Lecture Time Saved: 15 min

Summary Assessment: BioSecure AI converts classroom attendance from an error-prone administrative hurdle into an ambient, instantaneous, and verifiable digital workflow. Combining sub-millisecond local vector matching with autonomous biometric drift tracking, the platform provides educational institutions with a high-accuracy, friction-free attendance solution.